GPU sharing for multi-tenant inference

Quality assessment and data extraction

Group B | Interim evidence appendix | 16 October 2026

Nine provisional reports; 90 Q1-Q10 items; 34 registered outcome rows. Initial AI assessments only. Independent checks, final eligibility, report-to-study reconciliation and final PRISMA remain pending.

This appendix replaces the older seven-report handoff tables for current QA/extraction coverage. It is supporting material for the SLR, not the final synthesis report. Paper-by-paper tables must feed cross-study comparison, trade-off analysis and conditional recommendations.

What has been checked

Register keys, ten-item coverage, permitted scores, N/A denominators and totals were recalculated. Existing source scores and outcomes were preserved. This pass does not represent a new full-source appraisal or a real second reviewer.

What remains open

Twenty legacy QA items have section/figure anchors without printed PDF pages. Both provisional N/A judgments require calibration. All nine reports need independent checks; PipeSwitch scope and REEF conference/journal identity need resolution. Unchecked extraction fields remain not_extracted, rather than being converted to absent findings.

Measured review progress

Measure	Verified register state
Identification	4,189 occurrences; 111 applied duplicate links; 4,078 canonical records
Preliminary screening	2,487 active recommendations (61.0%); 1,591 awaiting first recommendation
Incoming AI opinions	1,831 preserved separately; 50 disagreements within 240 overlapping recommendations
Study selection	9 provisional report forms; final eligible reports and unique studies pending

Group B QA checklist and scoring

Authority: submitted protocol section 10, Table 8, printed pp11-12. Score 0 = absent/unclear, 1 = partly addressed, 2 = sufficiently supported. Full maximum 20. Use justified N/A only for a claim the paper does not make; subtract its maximum from the applicable denominator.

Normalized percentage = applicable score / applicable maximum x 100. Below 60% is a sensitivity analysis, not an exclusion gate. No initial report is below that line; calibration and source verification remain pending.

Item	Criterion	Requirements for score 2
Q1	Clarity of objective	Problem, proposed mechanism and evaluated questions are clear.
Q2	Technical description	Sharing primitive, configuration and implementation layer permit classification.
Q3	Experimental transparency	Setup, measurement procedure, repetitions or analytical method and result provenance are clear.
Q4	Hardware/workload description	GPU model/generation, software and concurrent inference workload/tenant characteristics are reported.
Q5	Baseline fairness	Relevant baseline and comparable workload, resources and measurements are documented or justified.
Q6	Utilization/throughput	Claimed benefit has defined metric, unit and method; otherwise alternate outcome is explained.
Q7	Latency measurement	Claimed latency specifies interval, load, statistic and method; claimed tail/SLO results state a quantile.
Q8	Multi-tenant interference	Concurrent tenants/co-tenants are tested and evidence supports interference conclusions.
Q9	Isolation evaluation	Claimed resource, security or fault isolation has a defined boundary and direct test or delimited evidence.
Q10	Replication/limits	Details/artifacts support informed replication; threats, limits or configuration dependence are discussed.

Study-by-criterion matrix

Report IDs are not unique-study counts. Scores remain preliminary; no human validation is implied. NA = justified provisional N/A, not a zero score.

ID	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	%
P001	2	2	1	1	1	2	NA	2	1	1	13/18	72.22
P002	2	2	2	1	1	2	2	2	1	2	17/20	85.0
P003	2	2	2	2	2	2	2	2	1	1	18/20	90.0
P004	2	2	2	1	2	2	2	2	1	1	17/20	85.0
P005	2	2	1	2	2	2	1	2	1	2	17/20	85.0
P006	2	2	1	2	1	2	2	2	NA	2	16/18	88.89
P007	2	2	2	2	1	2	2	2	2	2	19/20	95.0
P008	2	2	1	2	2	2	1	2	2	1	17/20	85.0
P009	2	2	1	1	1	2	2	2	1	2	16/20	80.0

Report key

Report	Title / initial QA status
GB-P001	VastPipe | AI_first_review;independent_check_pending;not_final
GB-P002	iGniter | AI_first_review;independent_check_pending;not_final
GB-P003	gpulets | AI_first_review;independent_check_pending;not_final
GB-P004	PipeSwitch | AI_first_review;independent_check_pending;not_final
GB-P005	REEF conference | AI_first_review;independent_check_pending;not_final
GB-P006	Paella: Low-latency Model Serving with Software-defined GPU Scheduling | AI_first_review;independent_check_pending;not_final
GB-P007	Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications | AI_first_review;independent_check_pending;not_final
GB-P008	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform | AI_first_review;independent_check_pending;not_final
GB-P009	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up | AI_first_review;independent_check_pending;not_final

N/A calibration

GB-P001 VastPipe Q7 and GB-P006 Paella Q9 are provisional N/A judgments. They are preserved, not ratified. The protocol requires a two-study calibration; no completed independent calibration is recorded.

GB-P001 | VastPipe

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Li-Xian Ma and Le-Ping Wang and En Shao and Rong-Yu Cao and Guang-Ming Tan / 2025 / Journal of Computer Science and Technology
DOI	10.1007/s11390-024-3773-5
Study linkage	pending_study_linkage
Source / anchor	s11390-024-3773-5.md | pp457-459, sections6.7-6.9, Figs14-16
Assessor / recorded date	Codex-GroupB | Item-level dates not populated in original register
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	13/18 (72.22%); AI first review
Required follow-up	Calibrate Q7 N/A against complete source; map section/figure anchors to original PDF pages.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Objective and local GPU portability question explicit	sections1,6.7
Q2	2/2	Tuned MPS quotas and CU masking described	sections6.7-6.8
Q3	1/2	GPU-specific repeats and variance unresolved in conversion	sections6.7-6.9
Q4	1/2	GPU/backend named; precise GPU loads and versions incomplete	sections6.7-6.8
Q5	1/2	Same hardware and no/default/tuned MPS; quotas incomplete	Fig14
Q6	2/2	Throughput ratios and baseline named in prose	Fig14 and prose
Q7	NA/0	GPU-portability extraction makes no latency claim; do not score NPU latency as GPU evidence	sections6.7-6.8
Q8	2/2	Multiple named co-located inference pairs evaluated	Figs14-16
Q9	1/2	Performance interference considered; no separate security/fault validation extracted	sections6.7-6.8
Q10	1/2	Stable-mix Q-table constraint; GPU replication settings incomplete	section6.9

Study-level extraction and missingness

Field	Registered value
mechanism	tuned/default MPS comparison; prior source check only
gpu hardware	NVIDIA A100 (existing source-check register)
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	not_extracted
models	VGG16+GoogLeNet (existing extraction rows)
tenant mix	not_extracted
batch load slo	not_extracted
baseline	No MPS / default MPS;exact quotas not extracted
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	not_extracted
tradeoffs	not_extracted

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O001
Metric / value / unit	Throughput ratio | 2.7 | x
Configuration	MPS tuned quotas | NVIDIA A100 | VGG16+GoogLeNet
Baseline	No MPS
Exact registered anchor	pp457-458 Fig14 and accompanying prose
Limits / interpretation	Batch/rate/quotas unresolved; not tail-latency evidence
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O002
Metric / value / unit	Throughput ratio | 1.5 | x
Configuration	MPS tuned quotas | NVIDIA A100 | VGG16+GoogLeNet
Baseline	Default MPS
Exact registered anchor	pp457-458 Fig14 and accompanying prose
Limits / interpretation	Same comparison family; not a universal average
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P002 | iGniter

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Fei Xu and Jianian Xu and Jiabin Chen and Li Chen and Ruitao Shang and Zhi Zhou and Fangming Liu / 2023 / IEEE Transactions on Parallel and Distributed Systems
DOI	10.1109/TPDS.2022.3232715
Study linkage	pending_study_linkage
Source / anchor	iGniter user-supplied Markdown | pp820-824, sections4.2,5.1-5.4, Figs13-17
Assessor / recorded date	Codex-GroupB | Item-level dates not populated in original register
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	17/20 (85.0%); AI first review
Required follow-up	Map section/figure anchors to original PDF pages; verify software/workload reporting.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Interference-aware inference provisioning objective explicit	abstract,section1
Q2	2/2	MPS allocation and shadow implementation described	section4.2
Q3	2/2	Model checks repeated three times with SD; metric windows specified	sections5.1-5.3
Q4	1/2	GPU/workloads/backend detailed; OS/driver unresolved	sections4.2,5.1
Q5	1/2	Adapted baselines and allocation/batch differences disclosed	section5.1
Q6	2/2	Cost, throughput, resource-allocation metrics distinguished	sections5.1,5.3
Q7	2/2	P99 windows, SLO and recovery timing explicit	section5.3 Fig17
Q8	2/2	Four co-located models and quota configuration specified	section5.2 Fig13
Q9	1/2	Performance interference measured; security/fault boundaries not tested in checked portion	sections5.2-5.3
Q10	2/2	Code, prediction limitations and PCIe caveat explicit	sections4.2,5.2

Study-level extraction and missingness

Field	Registered value
mechanism	MPS quota allocation and shadow process recovery;prior source check
gpu hardware	NVIDIA V100 (existing source-check register)
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	not_extracted
models	AlexNet;ResNet50;VGG19;SSD
tenant mix	not_extracted
batch load slo	batch3 for prediction check;W1 SLO10ms for recovery example
baseline	Measured latency for prediction error;10ms SLO for recovery
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	not_extracted
tradeoffs	not_extracted

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O003
Metric / value / unit	Prediction error range | 1.53-5.02 | percent
Configuration	MPS 25% per model | NVIDIA V100 | AlexNet/ResNet50/VGG19/SSD; batch3
Baseline	Measured local latency
Exact registered anchor	p822 section5.2 Fig13
Limits / interpretation	Prediction accuracy, not utilization or latency speedup
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O004
Metric / value / unit	Initial p99 latency | 15.6 | ms
Configuration	MPS shadow-process recovery | NVIDIA V100 | W1 with W5 and W11
Baseline	W1 latency SLO10ms
Exact registered anchor	p823 section5.3 Fig17 prose
Limits / interpretation	Transient violation precedes recovery; no unconditional guarantee
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P003 | gpulets

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Seungbeom Choi et al. / 2022 / USENIX ATC22
DOI	not resolved
Study linkage	pending_study_linkage
Source / anchor	https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf | pp201-203 local mechanism/Fig7; p207 Tables2-3;pp208-209 system methodology
Assessor / recorded date	Codex-GroupB | 2026-10-15T22:10:27+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	18/20 (90.0%); AI first review
Required follow-up	Recheck all criterion rationales against the exact source version.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Clear SLO-aware co-location objective	p199 abstract;section1
Q2	2/2	MPS partitions and temporal batching described	pp201-207 sections2-4
Q3	2/2	Poisson requests;median of3 repeats;throughput/SLO procedure	pp207-208 section5.1-5.2
Q4	2/2	Full GPU/CPU/OS/driver/runtime and model tables	p207 Tables2-3
Q5	2/2	Ported SBP comparator and Nexus incompatibility justified	p207 section5.1
Q6	2/2	Maximum throughput with99%SLO definition;system scale explicit	p208 section5.2
Q7	2/2	SLO99% procedure;local overhead CDF population explicit	p203 Fig7;p208 section5.2
Q8	2/2	Paired model interference and scheduler interference comparison	p203 Fig7;p208 Fig14
Q9	1/2	Performance interference checked;security/fault boundary not extracted	p203 Fig7;section5.2
Q10	1/2	Prototype and appendix setup available;replication not attempted	p207 Table2;appendixA

Study-level extraction and missingness

Field	Registered value
mechanism	MPS spatial partitions plus temporal batching; interference-aware gpulet scheduler
gpu hardware	RTX2080Ti11GB; two GPUs/server, two servers; local pair measurements separated
cpu memory	20-core Xeon E5-2630v4;192GB DRAM (Table2)
os kernel	Ubuntu18.04; kernel not extracted
driver runtime	NVIDIA440.64;CUDA10.2;PyTorch1.10 (p207 Table2)
models	GoogLeNet,LeNet,ResNet50,SSD-MobileNet,VGG16,MnasNet,MobileNetv2,DenseNet,BaseBERT (Table3)
tenant mix	paired local inferences; system scenarios contain several models
batch load slo	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
baseline	Solo latency for local Fig7; SBP and greedy best-fit for system scheduling
resource usage	not_extracted
scalability	8 logical backend servers with additional GPUs; system-level result (pp208-209)
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
advantages	not_extracted
limitations	Original Nexus not directly comparable; SBP ported; cross-model interference remains; no extracted security/fault test
tradeoffs	Partition and batch jointly selected; throughput numbers at system scale cannot be assigned to one GPU

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O006
Metric / value / unit	Distribution threshold:90% of measured pairs | 17.66 | percent overhead below threshold
Configuration	MPS spatial co-location | RTX2080Ti11GB | Paired model inferences
Baseline	Solo-running latency
Exact registered anchor	p203 Fig7
Limits / interpretation	Across-pair CDF, not request-p90/p99; no hostile-memory/fault test
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P004 | PipeSwitch

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Zhihao Bai;Zhen Zhang;Yibo Zhu;Xin Jin / 2020 / USENIX OSDI20
DOI	not resolved
Study linkage	pending_study_linkage
Source / anchor	https://www.usenix.org/system/files/osdi20-bai.pdf | p508 section6/6.1; p509 Tables3-4
Assessor / recorded date	Codex-GroupB | 2026-10-15T22:10:27+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	17/20 (85.0%); AI first review
Required follow-up	Resolve mixed training/inference scope using separable local inference measurements.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Fine-grained task-switching problem and design explicit	p499 abstract;section1
Q2	2/2	Controller,memory daemon,workers and pipeline described	pp504-508 sections4-5
Q3	2/2	100-run mean;min/max forFig6;setup described	p508 section6
Q4	1/2	Hardware/software specified;OS/kernel/driver not extracted	p508 section6
Q5	2/2	Same modified PyTorch;ready-model and swapping conditions explicit	p508 section6.1
Q6	2/2	Inference throughput and utilization denominator defined	p509 section6.1 Fig6
Q7	2/2	End-to-end overhead defined;mean and extrema identified	pp508-509 Table3
Q8	2/2	Training/inference co-location;inference metric separated	pp508-509 section6.1
Q9	1/2	Process/memory control described;security/fault testing not extracted	pp507-508 implementation
Q10	1/2	Versioned stack described;limitations documented;replication not attempted	p508 section6;section7

Study-level extraction and missingness

Field	Registered value
mechanism	Temporal task switching; pipelined model transfer; active/standby workers and shared memory daemon
gpu hardware	V10016GB PCIe3x16;T416GB PCIe3x8
cpu memory	V100 instance:8vCPU Xeon E5-2686v4/61GB;T4 instance:8vCPU Platinum8259CL/32GB (p508)
os kernel	not_extracted
driver runtime	CUDA10.1;PyTorch1.3.0;torchvision0.4.2;scipy1.3.2;driver not extracted
models	ResNet152;Inception_v3;Bert_base
tenant mix	one training task preempted for one inference task; separable inference latency
batch load slo	training32;inference8;100-run averages;Fig6 reports minimum/maximum (p508)
baseline	Ready model;stop-and-start;MPS using unified memory
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	Table3 total extra latency;Table4 first-layer startup overhead;kept separate
advantages	Reduced switching overhead for checked workloads; see extraction.csv
limitations	Mixed training/inference applicability;MPS benchmark swaps memory;modified PyTorch for all comparators;no extracted OS/driver/security/fault evidence
tradeoffs	Pipelined transfers and registered resident host models require runtime integration;inference-only multi-tenant relevance needs human eligibility check

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O005
Metric / value / unit	Additional end-to-end latency | 6.01 | ms
Configuration	Temporal switching | NVIDIA V10016GB | ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted; MPS overhead307.02ms
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O007
Metric / value / unit	Additional end-to-end latency | 307.02 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA V10016GB | ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O008
Metric / value / unit	Additional end-to-end latency | 5.40 | ms
Configuration	Temporal switching | NVIDIA V10016GB | Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O009
Metric / value / unit	Additional end-to-end latency | 232.25 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA V10016GB | Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O010
Metric / value / unit	Additional end-to-end latency | 10.27 | ms
Configuration	Temporal switching | NVIDIA V10016GB | Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O011
Metric / value / unit	Additional end-to-end latency | 204.52 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA V10016GB | Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O012
Metric / value / unit	Additional end-to-end latency | 5.57 | ms
Configuration	Temporal switching | NVIDIA T416GB | ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O013
Metric / value / unit	Additional end-to-end latency | 259.20 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA T416GB | ResNet152 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O014
Metric / value / unit	Additional end-to-end latency | 7.66 | ms
Configuration	Temporal switching | NVIDIA T416GB | Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O015
Metric / value / unit	Additional end-to-end latency | 193.05 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA T416GB | Inception_v3 inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O016
Metric / value / unit	Additional end-to-end latency | 34.56 | ms
Configuration	Temporal switching | NVIDIA T416GB | Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O017
Metric / value / unit	Additional end-to-end latency | 338.25 | ms
Configuration	NVIDIA MPS with unified-memory swapping | NVIDIA T416GB | Bert_base inference batch8 + training batch32
Baseline	Ready-model latency subtracted
Exact registered anchor	p509 Table3; setup p508 section6
Limits / interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P005 | REEF conference

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Mingcong Han;Hanze Zhang;Rong Chen;Haibo Chen / 2022 / USENIX OSDI22
DOI	not resolved
Study linkage	pending_study_linkage
Source / anchor	https://www.usenix.org/system/files/osdi22-han.pdf | p548 section7.1 Table2;p549 section7.2 Fig11;section8 idempotence limitation
Assessor / recorded date	Codex-GroupB | 2026-10-15T22:10:27+00:00
Eligibility	eligible_companion_candidate; final and independent decisions pending
QA total	17/20 (85.0%); AI first review
Required follow-up	Resolve conference/journal report relationship before study counting; do not substitute journal DOI.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Concurrent real-time/best-effort inference objective explicit	p539 abstract;section1
Q2	2/2	Reset preemption,queues and padding implementation described	sections3-6;pp547-548
Q3	1/2	Benchmark arrivals and baseline methods detailed;repeat/variance protocol not extracted	p548 section7.1
Q4	2/2	CPU,GPU,host RAM,OS,runtime,models andclients specified	p548 section7.1 Table2
Q5	2/2	RT-only and GPUStreams/SEQ conditions;extensions disclosed	p548 section7.1;p549 section7.3
Q6	2/2	Total RT+BE throughput and denominator explicit	p549 section7.2 Fig11
Q7	1/2	End-to-end latency and load specified;request percentile not extracted	pp548-549 Table2/Fig11
Q8	2/2	MultipleRT/BE clients and multiple mixes directly tested	p548 Table2;p549 section7.2
Q9	1/2	Performance isolation tested;no extracted memory/security/fault boundary test	pp548-549 section7.2
Q10	2/2	Artifact and idempotence/platform limitations explicit	p540 repository;section8;appendixA

Study-level extraction and missingness

Field	Registered value
mechanism	Reset-based preemption;idempotent inference kernels;dynamic kernel padding;runtime/driver changes
gpu hardware	AMD Radeon Instinct MI50:60CUs/16GB;REEF-N extension NVIDIA V100
cpu memory	Intel Corei7-10700:8cores;16GB host DRAM (p548)
os kernel	Ubuntu18.04;kernel not extracted
driver runtime	ROCm4.3.0;ApacheTVM0.8.0;NVIDIA extension CUDA10.2
models	ResNet152;DenseNet201;VGG19;Inceptionv3;DistilBERT
tenant mix	DISB A/B:1RT+1BE;C:1RT+5BE;D/E:5RT+5BE
batch load slo	Table2 RT100/220/100/20/20req/s;D uniform,E Poisson;BE closed-loop;batch not extracted
baseline	RT-Only;SEQ;GPUStreams;separate wait-based preemption microbenchmark
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	Checked workloads retain near RT-only latency while using idle capacity for BE inference
limitations	Reset requires idempotent kernels;model-pair padding incompatibilities;conference/journal report linkage open;not security/fault isolation
tradeoffs	DISB D/E throughput about25% below GPUStreams while limiting RT latency;implementation differs on closed-source GPUs

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O018
Metric / value / unit	Overall throughput ratio | 1.60 | x
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O019
Metric / value / unit	RT end-to-end latency overhead | 0.5 | percent
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	Described as negligible; not a p99 value or worst-case guarantee.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O020
Metric / value / unit	Overall throughput ratio | 1.14 | x
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	High RT load; differs from low-load DISB A.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O021
Metric / value / unit	RT end-to-end latency overhead | 1 | percent
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	About60 microseconds in prose; not a tail percentile.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O022
Metric / value / unit	RT end-to-end latency overhead | <1.5 | percent
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB C-E:multiple BE clients
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	Authors also state0.1ms; keep inequality and scenario range.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O023
Metric / value / unit	Overall throughput ratio | 3.00 | x
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB D:5 RT model clients20req/s uniform +5 BE model clients
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O024
Metric / value / unit	Overall throughput ratio | 2.96 | x
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | DISB E:5 RT model clients20req/s Poisson +5 BE model clients
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O025
Metric / value / unit	Overall throughput ratio | 7.7 | x
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | Apollo REAL trace about43 RTreq/s +5 BE model clients
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	Real-world trace mapped to representative DNN models; not production deployment measurement.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O026
Metric / value / unit	RT end-to-end latency overhead | <2 | percent
Configuration	Reset-based kernel preemption + dynamic kernel padding | AMD MI5016GB | Apollo REAL trace about43 RTreq/s +5 BE model clients
Baseline	RT-Only dedicated GPU
Exact registered anchor	p549 section7.2 Fig11 prose; setup p548 Table2
Limits / interpretation	Not a p99 latency claim or security/fault isolation test.
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P006 | Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Kelvin K.W. Ng; Henri Maxime Demoulin; Vincent Liu / 2023 / SOSP 2023
DOI	10.1145/3600006.3613163
Study linkage	GB-PS006
Source / anchor	https://kelvin-ng.github.io/assets/sosp2023-final224.pdf | printed pp604-606; PDF pp10-12; sections7.1-7.3
Assessor / recorded date	Codex-GroupB | 2026-10-16T10:00:46+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	16/18 (88.89%); AI first review
Required follow-up	Independently calibrate Q9 N/A: distinguish performance fairness from claimed resource/security/fault isolation.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Explicit goal	p595 section1
Q2	2/2	Mechanism classifiable	pp599-603 sections4-6
Q3	1/2	Repetition count unclear	p604 methodology
Q4	2/2	Versioned setup	p604 methodology
Q5	1/2	Wrapper caveat	p604 Table3/baselines
Q6	2/2	Defined throughput	pp604-605 Figs11-12
Q7	2/2	p99 specified	pp604-605 Figs11-12
Q8	2/2	Mixed jobs tested	p605 Figs12-13
Q9	NA/0	No security/fault or resource-isolation guarantee appraised; fairness is assessed under Q8; N/A provisional	pp606-607 discussion;scope provisional
Q10	2/2	Artifact and limits	pp606-607 section8

Study-level extraction and missingness

Field	Registered value
mechanism	Instrumented kernels; software scheduling
gpu hardware	Tesla T4
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Linux4.15.0;CUDA11.7;TVM0.10.0
models	Mixed inference; lognormal arrivals sigma1.5/2
tenant mix	Mixed inference; lognormal arrivals sigma1.5/2
batch load slo	not_extracted
baseline	CUDA-MS;Triton23.03;Paella ablations
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	Warmup excluded; TVM wrapper; MPS omitted from eight-model experiment
tradeoffs	Fairness versus short-job latency; instrumentation cost

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O027
Metric / value / unit	ResNet18 p99 reduction factor | up to3 | x
Configuration	Instrumented kernels; software scheduling | Tesla T4 | ResNet18+InceptionV3;lognormal sigma1.5/2
Baseline	CUDA-MS
Exact registered anchor	printed p604 section7.1/Fig12; PDF pp10-11
Limits / interpretation	Author-reported approximate maximum; not uniform across loads
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O028
Metric / value / unit	p90 instrumentation overhead | 6.6 | microseconds
Configuration	Instrumented kernels; software scheduling | Tesla T4 | Empty kernel;160 blocks;instrumentation with aggregation
Baseline	Uninstrumented kernel
Exact registered anchor	printed p606 section7.3/Fig15; PDF p12
Limits / interpretation	Microbenchmark; not inference request latency
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P007 | Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Foteini Strati; Xianzhe Ma; Ana Klimovic / 2024 / EuroSys 2024
DOI	10.1145/3627703.3629578
Study linkage	GB-PS007
Source / anchor	https://anakli.inf.ethz.ch/papers/orion_eurosys24.pdf | printed pp1082-1087; PDF pp8-13; sections6-7
Assessor / recorded date	Codex-GroupB | 2026-10-16T10:00:46+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	19/20 (95.0%); AI first review
Required follow-up	Recheck all criterion rationales against the exact source version.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Explicit goal	p1075 section1
Q2	2/2	Interception specified	pp1079-1082 sections4-5
Q3	2/2	Three repetitions	p1082 section6.1
Q4	2/2	Versioned setup	p1082 section6.1
Q5	1/2	Reimplemented comparator	p1083 baselines
Q6	2/2	Defined aggregate metric	p1086 section6.2.3
Q7	2/2	p99 and arrivals	p1086 section6.2.3
Q8	2/2	Collocations tested	p1086 section6.3
Q9	2/2	Trust boundary explicit	p1087 section7 Security
Q10	2/2	Artifact and limitations	p1087 section7;artifact appendix

Study-level extraction and missingness

Field	Registered value
mechanism	Interception; interference-aware kernel scheduling
gpu hardware	V10016GB;A10040GB extension
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	PyTorch1.12;Python3.9;CUDA10.2/11.3
models	High-priority plus best-effort inference;Poisson
tenant mix	High-priority plus best-effort inference;Poisson
batch load slo	not_extracted
baseline	Dedicated high-priority GPU;MPS;Streams;REEF-N policy
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	Trusted clients; no secure isolation; offline profiling; REEF-N reimplementation
tradeoffs	Tail latency versus best-effort progress

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O029
Metric / value / unit	Aggregate throughput ratio | up to7.3 | x
Configuration	Interception; interference-aware kernel scheduling | V10016GB;A10040GB extension | Two inference clients;Poisson;V100
Baseline	Dedicated high-priority inference
Exact registered anchor	printed p1086 section6.2.3/Fig12; PDF p12
Limits / interpretation	Maximum; includes best-effort work
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O030
Metric / value / unit	High-priority p99 overhead | 15 | percent
Configuration	Interception; interference-aware kernel scheduling | V10016GB;A10040GB extension | Two inference clients;Poisson;V100
Baseline	Dedicated high-priority inference
Exact registered anchor	printed p1086 section6.2.3/Fig12; PDF p12
Limits / interpretation	Authors summarize within15%; averaged across collocations, not worst-case guarantee
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P008 | GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Aditya Dhakal; Sameer G Kulkarni; K.K. Ramakrishnan / 2020 / SoCC 2020
DOI	10.1145/3419111.3421284
Study linkage	GB-PS008
Source / anchor	https://par.nsf.gov/servlets/purl/10299299 | printed pp499-501; PDF pp8-10; sections4.1-4.3
Assessor / recorded date	Codex-GroupB | 2026-10-16T10:02:57+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	17/20 (85.0%); AI first review
Required follow-up	Recheck all criterion rationales against the exact source version.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Explicit goal	p492 introduction
Q2	2/2	Sharing described	pp495-499 section3
Q3	1/2	Confidence intervals; counts unclear	p500 error bars;section4
Q4	2/2	Hardware and OS	pp499-500 testbed
Q5	2/2	Comparator scope stated	p501 section4.3
Q6	2/2	GUE defined	p494 GUE definition;p500 Table3
Q7	1/2	Tail statistic unclear	pp500-501 latency plots
Q8	2/2	Concurrent functions	p500 section4.2
Q9	2/2	Fixed quotas and performance-isolation tests; no stronger security claim inferred	pp496-498 resource controls;p502 performance isolation
Q10	1/2	Reproduction incomplete	pp499-504 implementation/evaluation/limits

Study-level extraction and missingness

Field	Registered value
mechanism	MPS allocation; adaptive batching
gpu hardware	One V10016GB
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Linux4.4.0-150;DPDK19.02;CNTK/PyTorch/TensorRT
models	Concurrent inference functions;25ms target
tenant mix	Concurrent inference functions;25ms target
batch load slo	not_extracted
baseline	Default MPS;MPS+SLAB;TensorRT19.02
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	GUE is composite, not hardware utilization percentage; no extracted tail percentile
tradeoffs	Batching/fairness versus individual latency

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O031
Metric / value / unit	GPU utilization efficacy GUE | 94.04 | composite score
Configuration	MPS allocation; adaptive batching | One V10016GB | Alexnet+ResNet50;25ms target
Baseline	Default MPS GUE38.82
Exact registered anchor	printed p500 Table3; PDF p9
Limits / interpretation	Not percent GPU utilization
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O032
Metric / value / unit	Signed latency improvement | -7.52 | percent
Configuration	MPS allocation; adaptive batching | One V10016GB | ResNet50;batch1;GPU-ready batch execution
Baseline	TensorRT server19.02
Exact registered anchor	printed p501 Table4; PDF p10
Limits / interpretation	Negative means worse latency; excludes transport overhead; single-model comparison
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

GB-P009 | Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Identity and assessment status

Field	Registered evidence
Authors / year / venue	Arpan Gujarati; Reza Karimi; Safya Alzayat; Wei Hao; Antoine Kaufmann; Ymir Vigfusson; Jonathan Mace / 2020 / USENIX OSDI 2020
DOI	No DOI resolved; retain official conference URL
Study linkage	GB-PS009
Source / anchor	https://people.mpi-sws.org/~antoinek/documents/20osdi_clockwork.pdf | printed pp451-452; author PDF pp9-10; section6.1/Fig5
Assessor / recorded date	Codex-GroupB | 2026-10-16T10:02:57+00:00
Eligibility	provisionally_eligible; final and independent decisions pending
QA total	16/20 (80.0%); AI first review
Required follow-up	Recheck all criterion rationales against the exact source version.

Detailed QA rationales

Initial assessor, status and evidence anchors come from evidence/quality.csv. Source-page precision is an audit flag, not a claim that this cycle independently verified the cited page.

Q	Score / max	Rationale	Source anchor
Q1	2/2	Explicit goal	p443 introduction
Q2	2/2	Scheduling described	pp448-451 sections4-5
Q3	1/2	Repetition count unclear	p451 section6 setup
Q4	1/2	Versions not extracted	p451 section6 setup
Q5	1/2	Hardware mismatch disclosed	p451 section6.1 mismatch
Q6	2/2	Goodput defined	p452 goodput definition
Q7	2/2	SLO and CDF	pp451-452 Fig5
Q8	2/2	Contending models	p451 section6.1
Q9	1/2	No fault guarantee	pp455-456 discussion
Q10	2/2	Artifact and limits	pp455-456 discussion;artifact appendix

Study-level extraction and missingness

Field	Registered value
mechanism	Predictable temporal scheduling
gpu hardware	One V10032GB in section6.1
cpu memory	not_extracted
os kernel	not_extracted
driver runtime	Not extracted
models	15 ResNet50 instances;16 closed-loop clients/model
tenant mix	15 ResNet50 instances;16 closed-loop clients/model
batch load slo	not_extracted
baseline	Clipper;INFaaS (different EC2 hardware)
resource usage	not_extracted
scalability	not_extracted
energy	not_extracted
reliability	not_extracted
security fault isolation	not_extracted
overhead	not_extracted
advantages	not_extracted
limitations	INFaaS comparison qualitative; fault tolerance unaddressed
tradeoffs	Predictability versus admission and concurrency

Registered quantitative outcomes

These are outcome rows, not separate studies. Preserve baseline, statistic, units and limitations. Not_extracted means unchecked; do not read it as not reported. Headline values require a real independent source check.

Outcome ID	GB-O033
Metric / value / unit	Clockwork goodput | approximately800 | requests/second
Configuration	Predictable temporal scheduling | One V10032GB in section6.1 | 15 ResNet50 instances;16 clients/model;500ms SLO
Baseline	INFaaS on nonidentical EC2 setup
Exact registered anchor	printed p452 section6.1/Fig5; author PDF p10
Limits / interpretation	Within-SLO successes; comparator qualitative; not total throughput
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending

Outcome ID	GB-O034
Metric / value / unit	Clockwork request latency | approximately300 | milliseconds
Configuration	Predictable temporal scheduling | One V10032GB in section6.1 | 15 ResNet50 instances;16 clients/model;500ms SLO
Baseline	No comparative ratio
Exact registered anchor	printed p452 section6.1/Fig5; author PDF p10
Limits / interpretation	Batch16 scheduling explanation; not a p99 value
Review status	Codex-GroupB | source_checked_first_reviewer; independent_check_pending; independent check pending